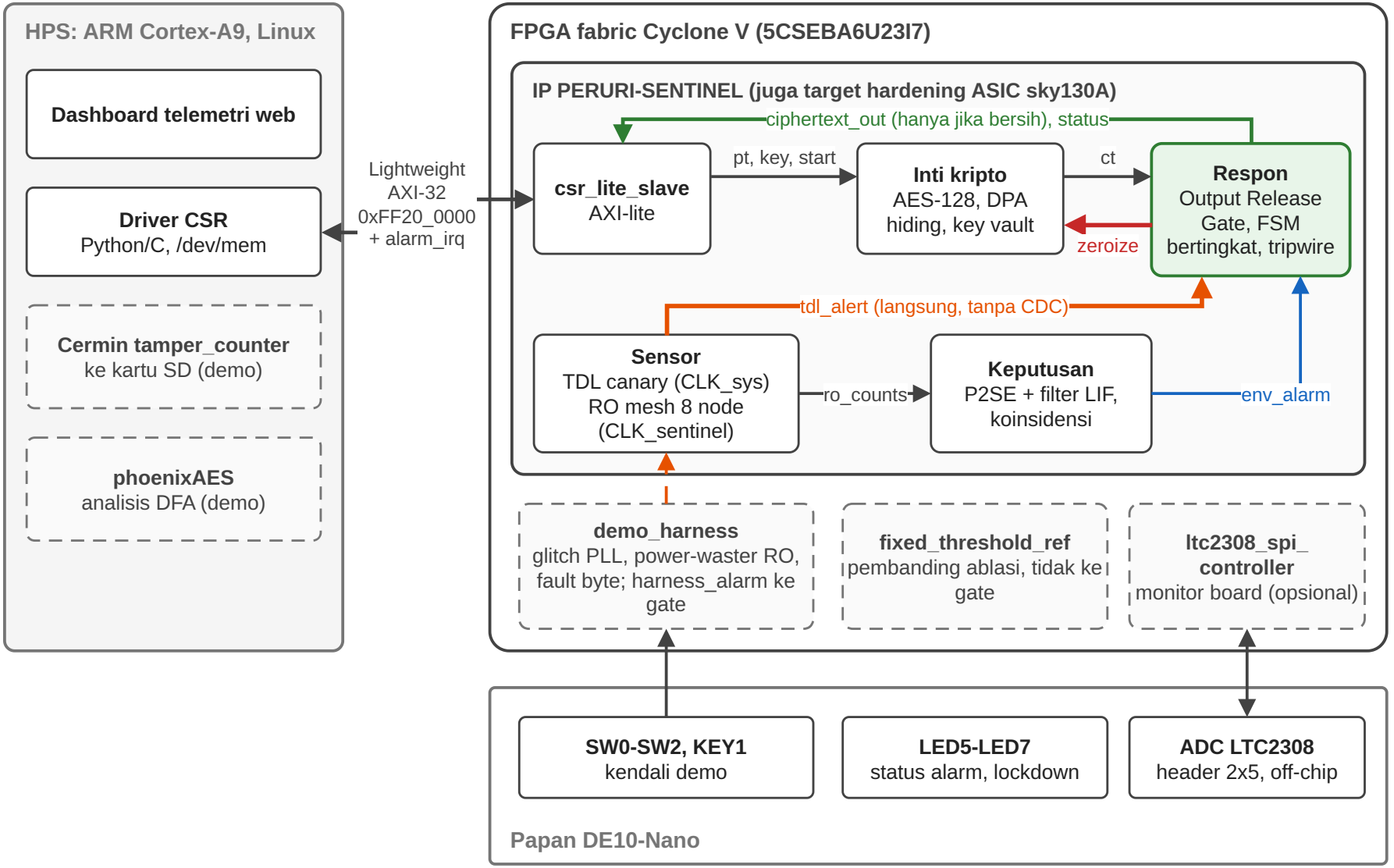
HPS: ARM Cortex-A9, Linux
Dashboard telemetri web
Driver CSR
Python/C, /dev/mem
Cermin tamper_counter
ke kartu SD (demo)
phoenixAES
analisis DFA (demo)
Lightweight
AXI-32
0xFF20_0000
+ alarm_irq
FPGA fabric Cyclone V (5CSEBA6U23I7)
IP PERURI-SENTINEL (juga target hardening ASIC sky130A)
ciphertext_out (hanya jika bersih), status
csr_lite_slave
AXI-lite
pt, key, start
Inti kripto
AES-128, DPA
hiding, key vault
ct
zeroize
Respon
Output Release
Gate, FSM
bertingkat, tripwire
tdl_alert (langsung, tanpa CDC)
Sensor
TDL canary (CLK_sys)
RO mesh 8 node
(CLK_sentinel)
ro_counts
Keputusan
P2SE + filter LIF,
koinsidensi
env_alarm
demo_harness
glitch PLL, power-waster RO,
fault byte; harness_alarm ke
gate
fixed_threshold_ref
pembanding ablasi, tidak ke
gate
ltc2308_spi_
controller
monitor board (opsional)
SW0-SW2, KEY1
kendali demo
LED5-LED7
status alarm, lockdown
ADC LTC2308
header 2x5, off-chip
Papan DE10-Nano
hanya build demo
jalur cepat TDL
alarm lambat
zeroize asinkron